

AKILESH A

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[LinkedIn](#) | [GitHub](#)

EXECUTIVE SUMMARY

Integrated MSc Software Systems Student with a strong foundation in data analysis, relational database architecture, predictive modeling, and intelligent software systems. Hands-on experience developing end-to-end data-driven applications, designing normalized SQL/PostgreSQL databases, and processing complex telemetry and geospatial datasets. Proficient in Python data science libraries (Pandas, NumPy), machine learning workflows, and REST API integration to transform raw data into actionable insights. Seeking Data Analytics, Data Science, Machine Learning, or Data Engineering roles to build scalable, high-impact data solutions.

EDUCATION

INTEGRATED MSC SOFTWARE SYSTEMS – Coimbatore Institute of Technology, Coimbatore

2024 – 2029

Cumulative GPA: 6.89 / 10.0 (Till 4th Semester)

ACADEMIC PROJECTS

VeloPulseAI – Predictive Telematics & Fleet Intelligence System [Python, Machine Learning, Scikit-Learn, Pandas, NumPy]

- **Telemetry Pipeline:** Engineered an AI-driven predictive maintenance pipeline ingesting multi-sensor vehicle telemetry to detect anomalous mechanical patterns prior to critical failure.
- **Data Preprocessing & EDA:** Engineered feature extraction and time-series preprocessing workflows in Python, isolating key degradation indicators and thermal/vibrational failure trends across simulated fleet data.
- **Health Scoring & Alerts:** Deployed predictive classification algorithms for real-time asset health scoring and automated dispatch of maintenance alerts.

RouteVeda – Intelligent Travel Planning Platform [PostgreSQL, PostGIS, React.js, Node.js, TypeScript, REST APIs] [GitHub](#)

- **Spatial Data Architecture:** Architected normalized relational PostgreSQL schemas with PostGIS spatial indexing for high-performance proximity queries and route optimization.
- **API Data Integration:** Integrated third-party REST APIs to ingest, validate, and synchronize dynamic multi-destination location datasets with sub-second response times.

Smart Hospital Resource Analytics System [MySQL, PHP, JavaScript, 8086 Assembly, HTML/CSS]

- **Resource Allocation Engine:** Built a healthcare resource analytics system to track and optimize critical patient intake, ICU/general bed capacity, oxygen inventory, and physician availability in real time.
- **Triage & Priority Analytics:** Implemented tri-tier priority triage classification (Critical, Serious, Normal) using rule-based decision logic to streamline emergency bed allocation and reduce patient wait times.
- **Relational Data Management:** Structured transactional MySQL database schemas to log admission/discharge events and generate aggregated ward utilization metrics for administrative decision-makers.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript
- **Data Analysis & ML:** Data Preprocessing, EDA, Feature Engineering, Data Cleaning, Statistical Analysis, Data Visualization
- **Python Data Stack:** Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Jupyter Notebook
- **Databases & Storage:** PostgreSQL, PostGIS, MySQL, MongoDB, SQLite, Oracle (PL/SQL), Firebase
- **Data Engineering & Integration:** REST API Data Integration, Data Pipelines, Relational Data Modeling, Database Design
- **AI:** Generative AI, Prompt Engineering, AI-Powered Application Design
- **Developer Tools:** Git, GitHub, VS Code, MySQL Workbench, EMU 8086
- **Soft Skills:** Teamwork & Collaboration, Time Management, Leadership, Willingness to Learn
- **Areas of Interest:** Data Analytics, Data Science, Machine Learning, Data Engineering, Business Intelligence, Predictive Analytics

ACHIEVEMENTS & CERTIFICATIONS

- **Smart India Hackathon 2026 (Internal Round):** Participated in internal college-level hackathon, collaborating in a multidisciplinary team to design and develop an innovative software solution under time constraints.
- **VeloPulseAI - Hackathon Project:** Contributed to developing an AI-driven predictive maintenance and telematics prototype for real-time vehicle fleet intelligence.
- **Introduction to Front-End Development:** Completed comprehensive training covering HTML, CSS, JavaScript, responsive web development fundamentals, and modern UI engineering.

LANGUAGES

Tamil | English |